

Gout: Diagnosis, Crystal Analysis & Differentials





1. Needle-shaped crystals

WHAT YOU SEE

Yellow needle crystals in thought-bubble with polarizer

WHAT IT ENCODES

Monosodium urate (MSU) crystals are needle-shaped and NEGATIVELY birefringent: yellow when parallel to the slow axis of the red compensator, blue when perpendicular. Synovial fluid polarized microscopy is the gold-standard diagnostic test for gout.

BOARD TRAP

CPPD (pseudogout) is rhomboid and POSITIVELY birefringent (blue parallel) — opposite color/shape to MSU.

2. Blue parallel = wrong

WHAT YOU SEE

'BLUE' crossed with X over needle

WHAT IT ENCODES

For MSU, blue when PARALLEL is incorrect — that pattern defines CPPD. MSU is yellow-parallel. Memorize: yelloW = neWative birefringence = urate.

BOARD TRAP

Confusing parallel-axis color is the classic exam trap separating gout from pseudogout.

3. Aspirate / arthrocentesis

WHAT YOU SEE

'ASPIRATE!' tag, needle

WHAT IT ENCODES

Joint aspiration with synovial fluid analysis is mandatory in acute mono-arthritis to confirm crystals AND exclude septic arthritis. WBC is typically 2,000–50,000 (inflammatory) in gout.

BOARD TRAP

Never assume gout on clinical grounds in a hot joint — septic arthritis must be excluded by aspiration first.

4. Septic / CPPD / cellulitis Ddx

WHAT YOU SEE

Top banner: SEPTIC, CPPD, PSORIASIS, CELLULITIS

WHAT IT ENCODES

Acute monoarthritis differential: septic arthritis (emergency), CPPD/pseudogout, psoriatic arthritis, cellulitis mimicking joint. Each is excluded by aspiration, crystal exam, and culture.

BOARD TRAP

Cellulitis overlying a joint can mimic gout — but cellulitis lacks intra-articular fluid and crystals.

5. Tophi — ear/olecranon

WHAT YOU SEE

Detective-board photos: ear helix, olecranon, finger

WHAT IT ENCODES

Tophi are chronic MSU deposits in cooler peripheral tissues: helix of ear, olecranon bursa, finger pads, Achilles tendon, first MTP. They signal long-standing untreated hyperuricemia and are an indication for urate-lowering therapy.

BOARD TRAP

Rheumatoid nodules also affect olecranon — but tophi are firm, chalky-white, and contain urate, not palisading granuloma.

6. Achilles / toe locations

WHAT YOU SEE

Achilles tendon and toe photos pinned

WHAT IT ENCODES

Classic gout/tophi sites: first MTP (podagra), midfoot, Achilles tendon insertion, toe. Lower-extremity cooler joints favor urate crystallization.

BOARD TRAP

Achilles enthesitis suggests spondyloarthropathy — but a chalky tophus there is gout, not enthesitis.

7. Acute attack (red/hot)

WHAT YOU SEE

Glowing red finger photo labeled ACUTE

WHAT IT ENCODES

Acute gout flare: sudden severe pain, erythema, warmth, swelling peaking in 12–24h. First-line acute therapy: NSAIDs, colchicine, or corticosteroids. Do NOT start or stop urate-lowering therapy during a flare.

BOARD TRAP

Starting allopurinol during an acute flare can prolong/worsen it — treat the flare first, then start ULT under prophylaxis.

8. Chalky white tophus

WHAT YOU SEE

'CHALKY WHITE' / 'PASTY' label, white paste on desk

WHAT IT ENCODES

Aspirated tophus material is chalky-white, pasty (toothpaste-like) due to dense MSU crystal aggregates. This gross appearance is pathognomonic of urate.

BOARD TRAP

Calcium deposits (CPPD/calcinosis) are also white on X-ray but synovial crystal microscopy distinguishes them.



9. Polarized microscopy

WHAT YOU SEE

'POLARIZED' label on microscope slide

WHAT IT ENCODES

Compensated polarized light microscopy of synovial fluid distinguishes MSU (needle, negative) from CPPD (rhomoid, positive). This is the definitive diagnostic modality.

BOARD TRAP

Plain light microscopy cannot determine birefringence — a polarizer with red compensator is required.

10. 5.4 mg/dL urate threshold

WHAT YOU SEE

'5.4' crossed crystal icon near flare

WHAT IT ENCODES

MSU saturates plasma around 6.8 mg/dL; urate-lowering therapy targets serum uric acid <6 mg/dL (<5 mg/dL with tophi). Serum urate may be normal/low DURING an acute flare, so a normal level does not exclude gout.

BOARD TRAP

A normal serum uric acid during the attack does NOT rule out gout — recheck after the flare resolves.

11. Flare trigger

WHAT YOU SEE

'FLARE' magnifier

WHAT IT ENCODES

Flares triggered by purine-rich food, alcohol (beer), dehydration, diuretics, and rapid urate shifts. IL-1β–driven inflammation underlies the attack.

BOARD TRAP

Loop/thiazide diuretics RAISE urate and precipitate gout — a common iatrogenic trigger to recognize on the boards.

12. IL-6 / cytokine pickpocket

WHAT YOU SEE

'IL-6 PICKPOCKET' bug icon

WHAT IT ENCODES

Gout inflammation is mediated by the NLRP3 inflammasome → IL-1β, with downstream IL-6 and TNF. This rationalizes IL-1 inhibitors (anakinra, canakinumab) for refractory flares.

BOARD TRAP

Anti-IL-6 (tocilizumab) is for RA/giant cell arteritis, not standard gout — the key cytokine target in gout is IL-1β.

13. Two crystals (DDx gun)

WHAT YOU SEE

'TWO CRYSTALS' vials with crystal gun

WHAT IT ENCODES

Two crystal arthropathies: MSU (gout, needle, negative birefringent) vs CPPD/calcium pyrophosphate (pseudogout, rhomboid, positive). Distinguishing them on aspiration directs management.

BOARD TRAP

CPPD chondrocalcinosis associates with hemochromatosis, hyperparathyroidism, hypomagnesemia — screen for these in pseudogout.

14. US / DECT / X-ray imaging

WHAT YOU SEE

Three imaging panels: US, DECT, X-RAY

WHAT IT ENCODES

Imaging clues: ultrasound 'double-contour sign' (urate on cartilage); dual-energy CT (DECT) color-codes urate deposits; X-ray shows late 'rat-bite' punched-out erosions with overhanging edges.

BOARD TRAP

DECT can be falsely positive from artifact; it supports but does not replace synovial crystal confirmation.

15. Culture (exclude septic)

WHAT YOU SEE

'CULTURE!' red banner with bacterium-gun

WHAT IT ENCODES

Always send synovial fluid for Gram stain and culture to exclude septic arthritis, which can coexist with crystals. Septic arthritis is a joint-destroying emergency requiring drainage and IV antibiotics.

BOARD TRAP

Finding crystals does NOT exclude concurrent infection — a septic joint can also contain crystals; culture is mandatory.